

Assignment 4: Exploration with AI/ML Engineering
Total 100 points

1. Getting setup:
 - a. Read the [Automated Line Identification and Notation Algorithm](#) (ALINA) paper.
 - b. fork the ALINA repository, and set it up locally on your computers. **(5 points)**
2. Run ALINA, where you will provide the ROI manually to the first frame in each video folder. (Use the data given in the repo). Play around with creating different sizes of ROI to understand how it warps a region into a bird's eye view. **(5 points)**
3. Exploring with experimentation and analysis
 - a. Explore different ML techniques (some examples: multimodal LLMs, VLMs, etc.) with which you can automate the ROI selection on the initial frame. **(20 points)**
 - b. Also, you may apply the automated approach on each frame (instead of only the first) and observe the difference in ROI selection between frames. Does it improve performance? **(5 points)**
4. Explore different ML techniques to identify the threshold of line marking colors (yellow and white taxiways), replacing the current manually set thresholding step. **(20 points)**

NOTE: 3a,b and 4 will be evaluated on creativity of leveraging state-of-the-art techniques and/or coming up with your own approach and the rigor of evaluation of different methods tested (**atleast 5**) and creating tables showing the performance with general metrics defined in ALINA (you may add more metrics as required). **(25 points)**

Submit the link to Github repository with README including procedures about how to execute and perform the experiments that you performed (the results should be reproducible since you are integrating ML components, so please use seeds). Provide the evidence through metrics evaluated, plotting graphs and video demonstrations of the end-to-end approach after including your best method from the different (atleast 5) techniques you came up with and evaluated.

Submit a report that integrates your experimentation and analysis which should include the following **(20 points)**:

1. Your designed ML pipeline

2. Experiments performed

3. Results and discussion

4. Limitations